

Multimodal progressive fusion and enhancement network for multimodal sentiment analysis

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ABSTRACT

Multimodal Sentiment Analysis (MSA) aims to identify a person's sentiment polarity or intensity by leveraging textual, acoustic, and visual modalities simultaneously, which is an important area of affective computing. Modeling sequential intra-modal information and enhancing cross-modal interactions are crucial to multimodal sentiment analysis. In this paper, we propose a **Multimodal Progressive Fusion and Enhancement Network (MPFEN)** to fully explore and model sequential intra-modal information and cross-modal interactions with a hybrid Mamba-Transformer architecture. The MPFEN consists of three key components, including the Multimodal Progressive Fusion Module (MPFM), the Multimodal Enhancement Module (MEM), and Modality Discrepancy Learning (MDL). The MPFM incorporates state-shared bidirectional Mamba (Bi-Mamba) and cross-modal attention mechanisms to facilitate information fusion across modalities progressively by explicitly modeling centralized cross-modal interaction. The MEM further refines the fused multimodal features with stacked Bi-Mamba-based multimodal enhancement layers. To fully leverage unimodal and multimodal fused features for sentiment prediction, the MDL module is designed to address the heterogeneity among modalities by modeling inter-modal discrepancies and output discrepancy scores that are aggregated with both unimodal and multimodal fused features to construct the final feature representation. Extensive experiments on 3 popular benchmarks, including CMU-MOSI, CMU-MOSEI, and CH-SIMS, demonstrate that our MPFEN achieves superior prediction performance and outperforms state-of-the-art baselines across different benchmarks.

1. Introduction

Multimodal Sentiment Analysis (MSA) is a crucial area of affective computing, focusing on predicting human sentiment states from multiple input data modalities simultaneously (Hu et al., 2024), including text, audio, and visual content. It has a wide range of scenario-based applications, particularly in healthcare services (Ezzameli & Mahersia, 2023), e-commerce (Wankhade et al., 2022), and recommendation systems (Darratz et al., 2025). Thus, it has attracted lots of attention in both the industrial community and the academic community. The key to MSA is learning a model to handle multiple modalities simultaneously and predict a sentiment state that can be a value or a class (Singh et al., 2024). Recently, most cutting-edge MSA methods have achieved impressive performances by representation learning and multimodal fusion. Representation learning-based methods (Hazarika et al., 2020; He et al., 2025; Hu et al., 2022; Li et al., 2023a,b; Mai et al., 2025; Yang et al., 2022, 2023a, 2024) focus on extracting refined modality semantics about human sentiment clues to improve the efficiency of multi-

modal fusion and relationship modeling. On the other hand, the multimodal fusion methods (Cai et al., 2025; Hou et al., 2024; Li et al., 2025a; Liu et al., 2018; Wu et al., 2023b; Xiong et al., 2025; Zhang et al., 2023) mainly explore the design of sophisticated fusion mechanisms to achieve an effective joint representation of multimodal data. The above-mentioned two types of methods demonstrate that reasonable sequential modeling and cross-modal interaction can significantly improve the performance of MSA.

However, existing methods still face two significant limitations. The first lies in how to capture long-range temporal modeling better. Convolutional neural networks (CNNs) and recurrent neural networks (RNNs) are efficient for sequential modeling, but CNNs are restricted by limited receptive fields, and RNNs suffer from vanishing gradient issues. Self-attention mechanisms like Transformers (Vaswani et al., 2017) are good at long-range sequence modeling, but they grapple with computational inefficiency. Mamba (Gu & Dao, 2023) architecture has recently emerged as a powerful alternative for sequence modeling. Built upon state space models (SSMs) (Gu et al., 2022, 2021b), Mamba combines a

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linear recurrence structure with hardware-efficient implementation, allowing it to capture long-range dependencies efficiently, often surpassing Transformers in specific areas, but there is still a certain gap with Transformers in long-range sequence modeling due to Transformers' advanced computational capabilities and proven effectiveness. In order to fully utilize their respective advantages to better model long-range temporal dependencies for the MSA task, effectively utilizing Mamba and Transformer modules remains an area that is worthy of exploration. The second lies in sub-optimal multimodal fusion. Different modalities are heterogeneous. Most existing approaches (Cai et al., 2025; Hou et al., 2024; Wu et al., 2023b; Zhang et al., 2023) directly perform cross-modal interaction without considering inter-modal heterogeneity and the primitive features. Such fusion may lead to sub-optimal multimodal fusion. To address this, we argue that effective fusion should first assess inter-modal differences by relying more on cross-modal features when heterogeneity is high, while appropriately retaining original unimodal characteristics when consistency is observed.

To address these limitations, we propose a Multimodal Progressive Fusion and Enhancement Network (MPFEN) to fully explore and model sequential intra-modal information and cross-modal interactions with a hybrid Mamba-Transformer architecture. The MPFEN consists of three key components, including the Multimodal Progressive Fusion Module (MPFM), the Multimodal Enhancement Module (MEM), and Modality Discrepancy Learning (MDL). Specifically, MPFEN first leverages pre-trained large models to encode text, audio, and visual inputs into deep, transferable unimodal representations. These representations are then passed into MPFM to perform coarse-to-fine feature alignment. The MPFM incorporates state-shared bidirectional Mamba (Bi-Mamba) and cross-modal attention mechanisms to facilitate information fusion across modalities progressively by explicitly modeling centralized cross-modal interaction. The state-shared Bi-Mamba is for local alignment and the cross-modal attention mechanism is to enable global semantic interaction. The fused multimodal features will be refined by the MEM, which consists of stacked Bi-Mamba-based multimodal enhancement layers. To fully leverage unimodal and multimodal fused features for sentiment prediction, the MDL module is designed to address the heterogeneity among modalities by modeling inter-modal discrepancies and output discrepancy scores that are aggregated with both unimodal and multimodal fused features to construct the final feature representation. To be specific, MDL computes the KL divergence between the output distributions of each modality's subnetwork and dynamically assesses their consistency. Then, it dynamically adjusts the weightings between unimodal representations and their fused multimodal counterparts before concatenating them for the final output. With this hierarchical fusion design, MPFEN can effectively combine powerful shallow features based on pre-trained models and newly extracted deep features to balance local modality-specific information and global multimodal information. Besides, it also adaptively manages modality discrepancies, leading to improved multimodal sentiment understanding. Extensive experiments on 3 popular benchmarks, including CMU-MOSI, CMU-MOSEI, and CH-SIMS, demonstrate that our MPFEN achieves superior prediction performance and outperforms state-of-the-art baselines across different benchmarks. In summary, our main contributions are summarized as follows:

- We propose a Multimodal Progressive Fusion and Enhancement Network (MPFEN) to fully explore and model sequential intra-modal information and cross-modal interactions with a hybrid Mamba-Transformer architecture.
- We introduce a Multimodal Progressive Fusion Module (MPFM) to perform hierarchical cross-modal alignment and interaction, and a Multimodal Enhancement Module (MEM) to further refine the fused features for improved sentiment understanding.
- To address the heterogeneity across modalities, we introduce a Modality Discrepancy Learning (MDL) module that dynamically adjusts the contribution of unimodal and multimodal features based on

their consistency to fully leverage unimodal features and multimodal fused features for sentiment prediction.

- Extensive experiments on 3 widely-used benchmark datasets, including CMU-MOSI, CMU-MOSEI, and CH-SIMS, demonstrate that our proposed MPFEN achieves superior prediction performance and outperforms state-of-the-art baselines across multiple evaluation metrics, validating its effectiveness and generalizability.

2. Related work

2.1. Multimodal sentiment analysis

Multimodal Sentiment Analysis (MSA) aims to predict individuals' emotional states by integrating information from various modalities, such as text, audio, and vision. In recent years, numerous approaches have been proposed to effectively model interactions between modalities. For instance, Tensor Fusion Network (Zadeh et al., 2017) employs Cartesian products to capture unimodal, bimodal, and trimodal interactions at different levels. Subsequently, Low-rank Multimodal Fusion (LMF) (Liu et al., 2018) improved upon TFN by performing parallel factorization of tensors and weights. It introduces modality-specific low-rank factors for multimodal fusion, significantly enhancing training efficiency. With the advent of the Transformer (Vaswani et al., 2017) architecture, many MSA models have adopted attention-based mechanisms for cross-modal alignment and fusion. Notable examples include the MulT model (Tsai et al., 2019) and the PMR model (Lv et al., 2021). Both of which leverage multimodal Transformers for inter-modal interaction. Hu et al. (2022), Wu et al. (2023b) further improved this line of research by incorporating pre-trained models to enhance the robustness and generalization of modality representations. Recent studies (Li & Li, 2025; Yang et al., 2023a, 2024) have adopted contrastive representation learning to improve the quality of multimodal representation learning. More recently, (Li et al., 2024a) achieves multimodal fusion based on the Mamba model by coupling multiple modality state chains. t-HNE (Li & Li, 2025) introduces a novel text-guided Hierarchical Noise Eliminator model, which effectively addresses noise caused by modality inconsistency in multimodal sentiment analysis through a two-stage denoising and feature recovery mechanism. EMOE (Fang et al., 2025) weights modalities per sample dynamically and uses unimodal distillation to preserve single-modality predictive ability. KAN-MCP (Luo et al., 2025) integrates a Kolmogorov-Arnold Network (KAN) for interpretable feature interaction modeling and a Multimodal Clean Pareto (MCPareto) framework to address modality imbalance and noise, enabling both explainable and robust multimodal sentiment analysis. MSamba (He et al., 2025) introduces the Mamba architecture to efficiently model both intra-modal and inter-modal sequential dependencies, demonstrating significant improvements in efficiency and performance.

2.2. State space models

State Space Models (SSMs), originally rooted in control theory, are mathematical frameworks designed to represent the input-output dynamics of time-varying systems. With the advancement of deep learning, SSMs have been increasingly adopted for sequence modeling, particularly for improving the efficiency and performance in handling long-range dependencies. Building upon early developments like LSSL (Gu et al., 2021b), subsequent enhancements such as S4 (Gu et al., 2021a) and S4D (Gu et al., 2022) have significantly advanced the capabilities of SSMs. Recently, Gu and Dao (2023) proposed Mamba, a novel language model that introduces a selective scanning mechanism combined with structured SSMs and hardware-aware parallelism. This design effectively addresses the quadratic complexity of Transformers, achieving linear-time computation without compromising performance. Inspired by Mamba's efficiency and success, several multimodal fusion approaches have emerged. For instance, MambaDFuse (Li et al.,

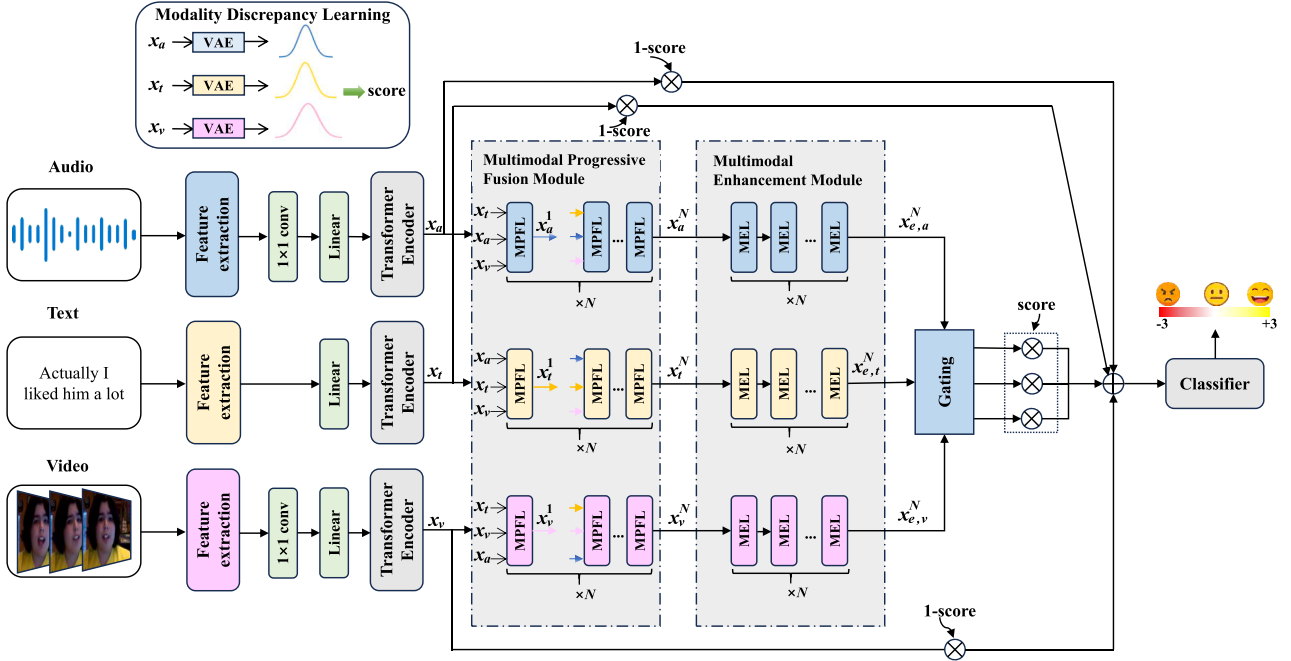


Fig. 1. The overall architecture of MPFEN. The model consists of three main components: MPFM performs progressive fusion across modalities. For each modality branch, MPFM consists of N stacked MPFL layers. Each MPFL is composed of a bidirectional Mamba module with shared transition matrix, an attention mechanism, and a Fusion-Enhancement Block, which together enable implicit-to-explicit cross-modal alignment and fusion. MEM further enhances the fused representations by refining the features produced by MPFM. MDL computes a modality-specific discrepancy score based on the Transformer-encoded features, which is used to adaptively control the contribution ratio between the original shallow features and the MEM-refined fused representations.

2024b) leverages unimodal Mamba encoders for cross-modal fusion, while AlignMamba (Li et al., 2025b) performs modality alignment prior to fusion within the Mamba framework. However, in the context of Multimodal Sentiment Analysis (MSA), the application of Mamba remains underexplored.

Building upon these advances, we propose a novel hybrid architecture that integrates Mamba with Transformer, aiming to combine their strengths of effective sequential modeling and fine-grained semantic alignment to enable more effective multimodal interaction and fusion.

3. Multimodal progressive fusion and enhancement network

3.1. Overview

Fig. 1 illustrates the overall architecture of our proposed Multimodal Progressive Fusion and Enhancement Network (MPFEN). The MPFEN takes text, audio, and video as input. Dedicated modality-specific encoders first process each modal input to generate unimodal sequence embeddings. Then, a modality-specific Transformer Encoder is applied to unimodal sequence embeddings of each modality to refine intra-modal sequential dependencies. These enhanced features are then passed through an MPFM, which stacks MPFLs that incorporate state-shared bidirectional Mamba (Bi-Mamba) and cross-modal attention mechanisms, to perform progressive feature alignment and fusion, enabling deep multimodal interaction centered on each modality. Subsequently, the fused representations obtained from the MPFM are further refined through an MEM, resulting in a set of multimodal sequence representations centered on each modality. A gating mechanism is then employed to choose effective information from multimodal sequence representations adaptively, resulting in a set of inter-modal features centered on each modality. Finally, to fully leverage intra-modal features and inter-modal features for MSA, the intra-modal features from the modality-specific Transformer Encoders and the inter-modal features are first modulated adaptively by the scores from the MDL module, and

then fed into Multi-Layer Perceptrons (MLPs) in a concatenated manner to generate the final prediction.

3.2. Intra-modal sequential information modeling

Given the multimodal input consisting of three modalities: language, audio, and video, precomputed sequential features $\mathbf{f}_m \in \mathbb{R}^{L^m \times D^m}$ are extracted with appropriate pre-trained models. $m \in \{t, a, v\}$ denotes the three modalities of language, audio, and video. L^m and D^m denote the sequence length and the feature dimension of the m modality features. For text modality, we adopt pre-trained RoBERTa (Liu et al., 2019) for extracting text sequential features $\mathbf{f}_t$. For audio modality, we utilize Whisper (Radford et al., 2023) to extract effective audio sequential features $\mathbf{f}_a$. For video modality, we adopt different pre-trained models since we found that the video quality varies greatly across different datasets. For the datasets with low video quality, like CMU-MOSI (Zadeh et al., 2016) and CMU-MOSEI (Zadeh et al., 2018), we extract the facial features as video features $\mathbf{f}_v$ using the Facet toolkit, reducing the impact of video quality on the utilization of features. For the datasets with high video quality, like CH-SIMS (Yu et al., 2020), we deploy the pre-trained model Timesformer (Bertasius et al., 2021) to extract video sequential features $\mathbf{f}_v$. After the extraction of precomputed sequential features, the audio features $\mathbf{f}_a$ and video features $\mathbf{f}_v$ are aligned in length with the text features $\mathbf{f}_t$ using a 1D convolution, addressing the temporal misalignment issue across modalities. Then, a linear projection is applied to all three modalities to unify their feature dimensions. Finally, a Transformer is employed to model the sequential dependencies within each modality, resulting in the processed representations $\mathbf{x}_m \in \mathbb{R}^{L \times D}$.

3.3. Deep centralized cross-modal interaction

Previous methods (Huang et al., 2020; Tsai et al., 2019) show the importance of cross-modal interaction in MSA since cross-modal interaction facilitates the learning of more effective representations by capturing the dependencies and correlations between modalities. To learn

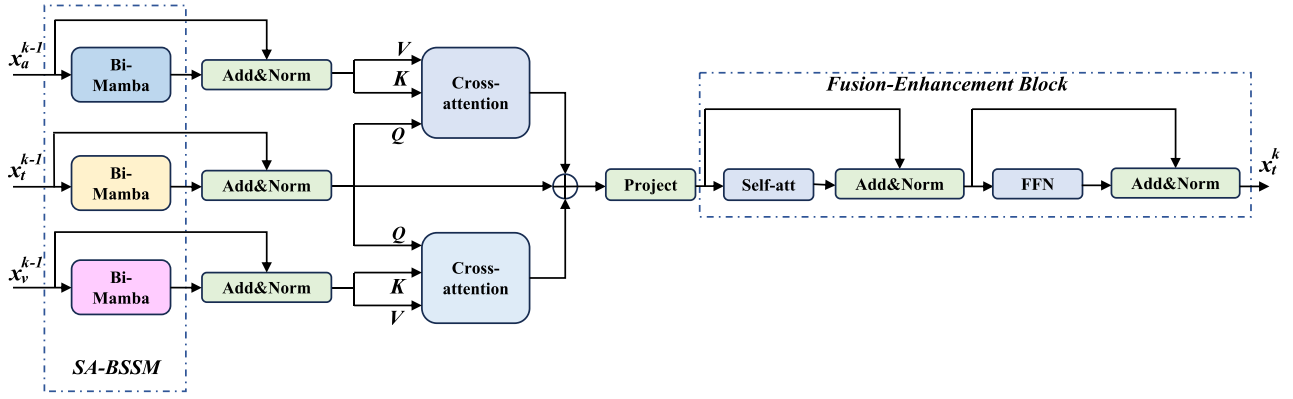


Fig. 2. The architecture of MPFL. The blue dashed box represents SA-BSSM, which denotes the Bidirectional Mamba module (Bi-Mamba) with a shared state transition matrix A .

more effective multimodal representations via cross-modal interaction, we deploy a deep centralized cross-modal interaction scheme to facilitate multimodal information alignment, fusion, and refinement progressively by explicitly modeling centralized cross-modal interaction with MPFM and MEM.

3.3.1. Multimodal progressive fusion module

The Multimodal Progressive Fusion Module (MPFM) is designed to align and fuse features from different modalities progressively, as illustrated in Fig. 1. It adopts a ternary symmetric structure where N Multimodal Progressive Fusion Layers (MPFLs) are stacked in each path. MPFM treats each modality as the central modality in turn and guides the deep cross-modal interaction and alignment of each modality with the other modalities progressively to fully exploit cross-modal complementary information. The structure of MPFL is illustrated in Fig. 2. In each MPFL, a state-shared Bidirectional Mamba is deployed to perform preliminary alignment across modalities by focusing on aligning local fine-grained features. Subsequently, a global attention mechanism based on cross-attention and self-attention is applied to capture high-level semantic relations, further refining the cross-modal alignment. This progressive alignment strategy gradually narrows the modality gap, enhances feature consistency, and facilitates semantically coherent multimodal fusion.

To describe how the MPFL operates, we take the text as the central modality for illustration. For the k th MPFL, it takes multimodal features $\mathbf{x}_t^{k-1}$, $\mathbf{x}_a^{k-1}$, and $\mathbf{x}_v^{k-1}$ as input. When k equals 1, $\mathbf{x}_t^{k-1} = \mathbf{x}_t$, $\mathbf{x}_a^{k-1} = \mathbf{x}_a$, and $\mathbf{x}_v^{k-1} = \mathbf{x}_v$. The MPFL is designed to capture information in the other two modalities that is relevant to the text modality, injecting their complementary semantic cues into the textual representation.

First, all features of the three modalities are processed by a shared-parameter Bidirectional Mamba (Bi-Mamba) block to obtain coarsely aligned features, followed by residual connection and layer normalization:

$$\mathbf{x}_m' = \text{LayerNorm}(\mathbf{x}_m^{k-1} + \text{SA-BSSM}(\mathbf{x}_m^{k-1})) \quad (1)$$

where the forward and backward state evolution processes of SA-BSSM, whose state matrices $\bar{A}$ and $\bar{A}$ are shared across the three modalities, is as follows:

$$\begin{aligned} \bar{h}_{m,t} &= \bar{A} \cdot \bar{h}_{m,t-1} + \bar{B}_m \cdot \mathbf{x}_{m,t}, \bar{y}_m = \bar{C}_m \cdot \bar{h}_{m,t} \\ \bar{h}_{m,t} &= \bar{A} \cdot \bar{h}_{m,t-1} + \bar{B}_m \cdot \mathbf{x}_{m,t}, \bar{y}_m = \bar{C}_m \cdot \bar{h}_{m,t} \end{aligned} \quad (2)$$

The right arrow represents the forward pass, and the left arrow represents the backward pass. Then, the text modality features serve as an anchor to apply cross-attention to the other two modalities' features. Taking text as the anchor and audio as the auxiliary modality as an example, the cross-modal representations are formulated as follows:

Given the text features $\mathbf{x}_t' \in \mathbb{R}^{l_t \times d_t}$ and the audio feature representations $\mathbf{x}_a' \in \mathbb{R}^{l_a \times d_a}$, the expressions for the query $Q_t \in \mathbb{R}^{l_t \times d_k}$, key $K_a \in$

$\mathbb{R}^{l_a \times d_k}$, and value $V_a \in \mathbb{R}^{l_a \times d_k}$ in the cross-modal attention from audio to text are:

$$\begin{aligned} Q_t &= \mathbf{x}_t' W_Q, \\ K_a &= \mathbf{x}_a' W_K, \\ V_a &= \mathbf{x}_a' W_V, \end{aligned} \quad (3)$$

where $W_Q \in \mathbb{R}^{d_t \times d_k}$, $W_K \in \mathbb{R}^{d_a \times d_k}$, and $W_V \in \mathbb{R}^{d_a \times d_k}$ are learnable weight matrices. Following the standard approach (Vaswani et al., 2017), the cross-modal attention operation from the auxiliary audio modality to the text anchor produces the enhanced text-based embeddings $\mathbf{x}_{a \rightarrow t} \in \mathbb{R}^{l_t \times d_t}$ as:

$$\mathbf{x}_{a \rightarrow t} = \text{softmax}\left(\frac{Q_t K_a^T}{\sqrt{d_k}}\right) V_a \quad (4)$$

Therefore, the entire process can be described as follows:

$$\begin{aligned} \mathbf{x}_{a \rightarrow t} &= \text{Cross-Attn}(\mathbf{x}_t', \mathbf{x}_a', \mathbf{x}_a') \\ \mathbf{x}_{v \rightarrow t} &= \text{Cross-Attn}(\mathbf{x}_t', \mathbf{x}_v', \mathbf{x}_v') \end{aligned} \quad (5)$$

Subsequently, the features $\mathbf{x}_t'$, $\mathbf{x}_{a \rightarrow t}$, and $\mathbf{x}_{v \rightarrow t}$ are concatenated and passed through a projection layer, which is a Linear layer, followed by a Transformer-like sub-layer structure, referred to as Fusion Enhancement Block (FEB), for feature enhancement and obtaining the output $\mathbf{x}_t^k$:

$$\mathbf{x}_t^k = \text{FEB}(\text{Project}(\text{Concat}(\mathbf{x}_t', \mathbf{x}_{a \rightarrow t}, \mathbf{x}_{v \rightarrow t}))) \quad (6)$$

For simplicity, the k th MPFL for each modality can be summarized as follows:

$$\begin{aligned} \mathbf{x}_t^k &= \text{MPFL}(\mathbf{x}_t^{k-1}, \mathbf{x}_a^{k-1}, \mathbf{x}_v^{k-1}) \\ \mathbf{x}_a^k &= \text{MPFL}(\mathbf{x}_a^{k-1}, \mathbf{x}_t^{k-1}, \mathbf{x}_v^{k-1}) \\ \mathbf{x}_v^k &= \text{MPFL}(\mathbf{x}_v^{k-1}, \mathbf{x}_t^{k-1}, \mathbf{x}_a^{k-1}) \end{aligned} \quad (7)$$

By stacking N MPFLs, the MPFM takes $\mathbf{x}_t$, $\mathbf{x}_a$, and $\mathbf{x}_v$ as input and output three intermediate multimodal features $\mathbf{x}_t^N$, $\mathbf{x}_a^N$, and $\mathbf{x}_v^N$:

$$\mathbf{x}_t^N, \mathbf{x}_a^N, \mathbf{x}_v^N = \text{MPFM}(\mathbf{x}_t, \mathbf{x}_a, \mathbf{x}_v) \quad (8)$$

3.3.2. Multimodal enhancement module

To refine and facilitate contextual modeling of the fused multimodal features for learning a more effective multimodal representation, we introduce a Multimodal Enhancement Module (MEM) comprising N Multimodal Enhancement Layers (MEL). As shown in Fig. 3, drawing inspiration from the Transformer architecture, each MEL integrates a bidirectional Mamba block followed by a residual feed-forward network (FFN) that enhances nonlinear positional interactions, thereby improving the

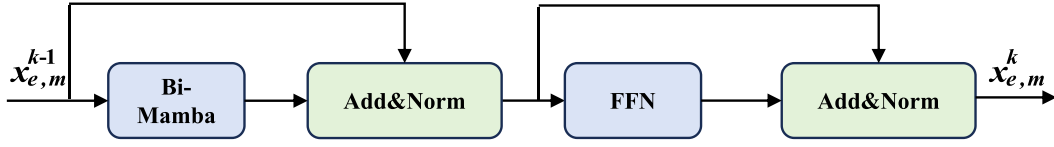


Fig. 3. The architecture design of MEL, which consists of Bi-Mamba, a feed-forward network (FFN), and residual connections. The fused modalities are further refined by Bi-Mamba and FFN to enhance their joint representation.

overall representational power. k th MEL for the m modality can be formalized as follows:

$$\begin{aligned} \mathbf{x}_{e,m}' &= \text{LayerNorm}(\mathbf{x}_{e,m}^{k-1} + \text{BiMamba}(\mathbf{x}_{e,m}^{k-1})) \\ \mathbf{x}_{e,m}^k &= \text{LayerNorm}(\mathbf{x}_{e,m}' + \text{FFN}(\mathbf{x}_{e,m}')) \end{aligned} \quad (9)$$

For simplicity, it can be formalized as follows:

$$\mathbf{x}_{e,m}^k = \text{MEL}(\mathbf{x}_{e,m}^{k-1}) \quad (10)$$

When $k = 1$, $\mathbf{x}_{e,m}^{k-1}$ is the fused representation $\mathbf{x}_m^N$. By stacking N MELs, the MEM takes $\mathbf{x}_t^N$, $\mathbf{x}_a^N$, and $\mathbf{x}_v^N$ as input and output three refined multimodal features $\mathbf{x}_{e,t}^N$, $\mathbf{x}_{e,a}^N$, and $\mathbf{x}_{e,v}^N$:

$$\mathbf{x}_{e,t}^N, \mathbf{x}_{e,a}^N, \mathbf{x}_{e,v}^N = \text{MEM}(\mathbf{x}_t^N, \mathbf{x}_a^N, \mathbf{x}_v^N) \quad (11)$$

3.4. Modality discrepancy learning

Multimodal learning can often improve the network performance by aggregating more information sources and constructing more robust feature representations. However, when emotional cues across modalities are inconsistent, the effectiveness of multimodal fusion may degrade. Following previous work (Wang et al., 2024), we leverage unimodal features and multimodal fused features jointly with modality discrepancy learning (MDL) to address this issue for MSA. Specifically, when the inter-modal inconsistency is low, the model relies more on the unimodal features. Otherwise, the model relies more on the multimodal fused features with cross-modal correlations. To quantify the modality inconsistency, we compute a heterogeneity score by measuring the pairwise Kullback-Leibler (KL) divergence among the three modalities after the intra-modal sequential information modeling. To be specific, we first employ a Variational Autoencoder (VAE) for each modality to obtain the posterior distributions, from which latent variables $q(z_t|\mathbf{x}_t)$, $q(z_a|\mathbf{x}_a)$, and $q(z_v|\mathbf{x}_v)$ are sampled by reparameterization as follows:

$$q(z_m|\mathbf{x}_m) = \mathcal{N}(z_m|\mu(\mathbf{x}_m), \sigma(\mathbf{x}_m)) \quad (12)$$

Then, the heterogeneity score is computed as the average pairwise KL divergence among the three distributions as follows:

$$\text{score} = \text{Sigmoid}\left(\frac{s_{ta} + s_{at} + s_{tv} + s_{vt} + s_{av} + s_{va}}{6}\right) \quad (13)$$

where $s_{ij} = D_{KL}(q(z_i|\mathbf{x}_i)||q(z_j|\mathbf{x}_j))$, $i \in \{t, a, v\}$, $j \in \{t, a, v\}$ and $i \neq j$. D_{KL} is the KL divergence. When the heterogeneity score is close to 1.0, the contribution of unimodal features is discarded, and the final representation is dominated by the deeply fused outputs from the MPFM and MEM modules, since the emotional cues across modalities are consistent and can be fused effectively. Otherwise, our model will consider the contribution of unimodal features more when fusing features to obtain the final representation. In this way, we can prevent conflicting raw modality information from degrading fusion quality when inter-modal discrepancy is high.

To better aggregate the unimodal feature $\mathbf{x}_m$ with the fused multimodal representation $\mathbf{x}_{e,m}^N$, we purify the cross-modal feature $\mathbf{x}_{e,m}^N$ since it may contain redundant information. Hence, we adopt a gating network composed of multi-layer perceptrons (MLPs) to adaptively weight the cross-modal features $\mathbf{x}_{e,m}^N$, thereby reducing redundant information and preventing performance degradation caused by information overlap. With the heterogeneity score, the final feature representation f

given by MDL can be modeled as follows:

$$\begin{aligned} \tilde{\mathbf{x}}_{e,m} &= \text{Gating}(\text{Pool}(\mathbf{x}_{e,m}^N) * \text{Pool}(\mathbf{x}_m^N) * \text{score}) \\ \tilde{\mathbf{x}}_m &= \text{Pool}(\mathbf{x}_m) * (1 - \text{score}) \\ f &= \text{Concat}(\tilde{\mathbf{x}}_t, \tilde{\mathbf{x}}_a, \tilde{\mathbf{x}}_v, \tilde{\mathbf{x}}_{e,t}, \tilde{\mathbf{x}}_{e,a}, \tilde{\mathbf{x}}_{e,v}) \end{aligned} \quad (14)$$

where $\text{Pool}(\cdot)$ denotes AdaptiveMaxPool.

3.5. Context modeling

In sentiment analysis tasks, context modeling can not only enhance the model's understanding of emotion-related words but also improve overall semantic consistency and prediction accuracy. It serves as a key mechanism for achieving high-quality sentiment analysis. To this end, we introduce contextual information $\mathbf{x}_m^c$ to enhance the sentiment analysis of the current sample. We process the context and the current input separately, and then concatenate their output representations to obtain a unified feature representation. To be specific, we set the context window for textual information to 2, and for audio and visual information to 1. Specifically, we separately process the current sample information and the contextual information. The modalities of the current sample need to be encoded and output as three types of modality encodings before passing through the MPFM. Similarly, the three modalities of the contextual information also undergo the same encoding process. Meanwhile, the encoded information of the contextual information is further processed through subsequent MPFM and MEM to produce three outputs. By connecting these six outputs from the contextual information with the 6 outputs from the current sample, we can obtain the features enriched with contextual information for the current sample.

3.6. Sentiment prediction & training objective

With the final feature representation f , we obtain the sentiment prediction by passing f through a multilayer perceptron (MLP) to produce an output y_f . Besides, we also learn to predict based on the unimodal features $\mathbf{x}_m$ and the final representation f^c enhanced with contextual features $\mathbf{x}_m^c$ for enhancing the model's capability, we also utilize the context to pass through the subnets of the three modalities, and obtain the following outputs: y_m , y_m^c , and y_{fc} . The L1 loss function is employed to compute the losses between the ground-truth y and predicted outputs y_f/y_{fc} , y_t , y_a , y_v , and y_m^c . We denote these losses as $\mathcal{L}_{multi}$, $\mathcal{L}_t$, $\mathcal{L}_a$, $\mathcal{L}_v$, and $\mathcal{L}_c$, respectively. Therefore, the overall learning objective is as follows:

$$\mathcal{L} = \mathcal{L}_{multi} + \alpha(\mathcal{L}_t + \mathcal{L}_a + \mathcal{L}_v) + \beta\mathcal{L}_c \quad (15)$$

where α and β denote the weighting factors for unimodal loss and contextual loss, both constrained within the range $[0,1]$.

4. Experiments

4.1. Datasets, metrics, and baselines

To evaluate the effectiveness of our proposed Multimodal-Mamba model, we conduct experiments on Multimodal Sentiment Analysis (MSA) tasks. For the MSA task, we use commonly adopted English datasets MOSI (Zadeh et al., 2016) and MOSEI (Zadeh et al., 2018),

Table 1
Data statistics of CMU-MOSI, CMU-MOSEI, and CH-SIMS.

Dataset	Train	Valid	Test	Total	Language
CMU-MOSI	1284	229	686	2199	English
CMU-MOSEI	16326	1871	4659	22856	English
CH-SIMS	1368	456	457	2281	Chinese

as well as a widely-used Chinese dataset CH-SIMS (Yu et al., 2020). A summary of the datasets is provided in Table 1.

Following prior works (Feng et al., 2024; He et al., 2025; Yang et al., 2024), we adopt Mean Absolute Error (MAE), Pearson correlation Coefficient (Corr), seven-class classification accuracy (Acc-7), binary classification accuracy (Acc-2), and F1 score as evaluation metrics. Specifically, Acc-2 and the F1 score are calculated based on a non-positive/positive standard. We compare MPFEN with 18 state-of-the-art baselines, including **LF-DNN** (Williams et al., 2018), **MFM** (Tsai et al., 2018), **LMF** (Liu et al., 2018), **TFN** (Zadeh et al., 2017), **MuLT** (Tsai et al., 2019), **MISA** (Hazarika et al., 2020), **MAG-BERT** (Rahman et al., 2020), **HyCon** (Mai et al., 2022), **Self-MM** (Yu et al., 2021), **ConFEDE** (Yang et al., 2023a), **DMD** (Li et al., 2023b), **ALMT** (Zhang et al., 2023), **DBF** (Wu et al., 2023a), **UniMSE** (Hu et al., 2022), **HyDiscGAN** (Wu et al., 2024), **MCL-MCF** (Fan et al., 2024), **CLGSI** (Yang et al., 2024), and **MSAmba** (He et al., 2025). The brief introduction of these baselines is listed as follows.

- **LF-DNN** employs input-level multimodal fusion that integrates audio, visual, and textual features for emotion and intensity recognition. The fused features are processed by Bidirectional Long Short-Term Memory deep neural networks to capture temporal dependencies across modalities. The system jointly performs multi-class classification and regression.
- **MFM** proposes a Multimodal Factorization Model that jointly optimizes generative and discriminative objectives to learn robust multimodal representations. The model factorizes features into two independent components: multimodal discriminative factors that capture shared information across modalities for prediction, and modality-specific generative factors that retain reconstruction information unique to each modality.
- **LMF** proposes a low-rank multimodal fusion framework that integrates audio, visual, and textual features at the input level. By decomposing high-dimensional tensors into low-rank factors, it efficiently captures cross-modal interactions while reducing computational and memory costs. This approach effectively models multimodal dependencies through early-stage tensor fusion.
- **TFN** introduces a tensor-based fusion network that explicitly models intra- and inter-modal dynamics among different modalities.
- **MuLT** proposes a multimodal transformer architecture that translates one modality to another using directional pairwise cross-attention.
- **MISA** designs a framework that separates modality-invariant and modality-specific representations to capture shared and unique information.
- **MAG-BERT** introduces a Multimodal Adaptation Gate designed to extend pre-trained language models such as BERT and XLNet for multimodal sentiment analysis. The gate integrates visual and acoustic information by generating modality-conditioned shifts to the internal representations of these models during fine-tuning.
- **HyCon** proposes a hybrid contrastive learning framework for multimodal sentiment analysis that jointly performs intra-, inter-, and semi-contrastive learning across modalities.
- **Self-MM** designs a self-supervised learning framework to rectify labels for unimodality.
- **ConFEDE** is a unified framework for multimodal sentiment analysis that combines contrastive representation learning with contrastive feature decomposition. It factorizes text, audio, and video modalities

ties into similarity and dissimilarity features, guided by contrastive relations centered on text.

- **DMD** proposes a decoupled multimodal distillation framework for emotion recognition, which separates each modality into modality-irrelevant and modality-exclusive representations. A graph distillation unit is applied to each part, enabling flexible and adaptive cross-modal knowledge transfer with learnable distillation weights.
- **ALMT** introduces an Adaptive Language-guided Multimodal Transformer for sentiment analysis, which leverages an Adaptive Hypermodality Learning module to suppress modality-irrelevant and conflicting information from audio and visual features under the guidance of language.
- **DBF** introduces a denoising bottleneck fusion model for video multimodal tasks, integrating visual, audio, and textual features while filtering noise and redundancy.
- **UniMSE** unifies multimodal sentiment analysis and emotion recognition in conversation by integrating features, labels, and models.
- **HyDiscGAN** introduces a hybrid distributed cross-modality cGAN framework for multimodal sentiment analysis, focusing on modality-specific privacy preservation. The model generates de-identified audio and visual features conditioned on shareable textual data, ensuring privacy while maintaining cross-modal alignment.
- **MCL-MCF** introduces a progressive multimodal fusion framework for sentiment analysis that mitigates modality heterogeneity through multi-level contrastive learning.
- **CLGSI** introduces a contrastive learning framework for multimodal sentiment analysis guided by sentiment intensity. Positive and negative sample pairs are selected based on intensity differences and weighted accordingly, while a Global-Local-Fine-Knowledge fusion mechanism integrates common and modality-specific features.
- **MSAmba** is a hybrid Mamba-based model for multimodal sentiment analysis, combining intra-modal sequential modeling and cross-modal interaction via Mamba and attention.

4.2. Implementation details

To optimize our model, we set the batch size to 16 for CMU-MOSI and CMU-MOSEI, with hyperparameters $\alpha = 0.3$ and $\beta = 0.1$. For CH-SIMS, the batch size is set to 8, with $\alpha = 0.3$ and $\beta = 0$. The MPFM and MEM in our model are configured with 2 layers. The learning rates were set to 1×10^{-5} for CMU-MOSI, 5×10^{-6} for CMU-MOSEI, and 1×10^{-5} for CH-SIMS. We train the model using the AdamW (Loshchilov & Hutter, 2017) optimizer and apply a cosine learning rate scheduler for dynamic learning rate adjustment. Gradient clipping is set to 20 to prevent gradient explosion. During training, we train our model for 100 epochs with an early stopping strategy that has a patience value of 5. All experiments are conducted on 2 NVIDIA RTX 4090D GPUs. We run experiments three times with 3 different random seeds: [1111, 2222, 3333] and report the average results and the best results in the Tables 2–4. In the ablation study, we report the best results for comparison. We also design another version of MPFEN with small sizes for a fair comparison. We name this model as **MPFEN_{small}**, where the model configuration is set to a hidden layer size of 64, and both the MPFM and MEM modules are configured with 1 layer.

4.3. Comparison to state-of-the-art methods

As shown in Tables 2–4, our model **MPFEN^{best} + c** outperforms the baselines across all three datasets. Compared with the state-of-the-art MSAmba, our **MPFEN^{best} + c** demonstrates a performance advantage on the CMU-MOSI dataset, surpassing it by 3.27%, 3.29%, and 0.77% in Acc-2, F1 score, and Acc-7, respectively. On the CMU-MOSEI dataset, it also outperforms MSAmba by 2.22%, 2.07%, and 0.95% in Acc-2, F1 score, and Acc-7, respectively. Furthermore, on the more challenging Chinese dataset CH-SIMS, (**MPFEN**)^{best} + c achieves improvements

of 2.38%, 2.96%, and 2.94% in Acc-2, F1 score, and Acc-3, respectively. These results demonstrate that by integrating Mamba and cross-attention mechanisms, our model can capture cross-modal information more effectively and achieve better multimodal fusion for MSA. Besides, even without the help of context, our $\text{MPFEN}_{\text{small}}$ and MPFEN still outperform the baselines across all three datasets. These results fully show the superiority of our method.

4.4. Ablation study on different components

We conducted ablation studies to evaluate the effectiveness of each component in our model, as shown in Table 5. The results demonstrate that removing the MPFM leads to the most significant performance drop, with Acc-2 decreasing by 1.83%, underscoring the critical importance of cross-modal interaction in multimodal learning. In addition, removing the MEM also results in noticeable degradation, indicating that the fused features still contain latent semantic cues that can be further exploited. MEM plays an essential role in refining these post-fusion representations. Moreover, the removal of the MDL leads to decreases of 0.79% and 1.41% in Acc-2 and Acc-7, respectively, compared with the complete MPFEN. These results suggest that MDL is beneficial in alleviating the adverse effects of semantic inconsistency across modalities.

Our MPFEN adopts a ternary symmetric interaction fusion strategy, which may introduce redundant information in the final representation. To address this, we incorporate a Gating mechanism. The ablation results show that excluding this component also causes performance degradation, validating its role in effective information filtering.

Finally, we study the effects of the loss terms L_{uni} and L_c . The results show that removing L_{uni} leads to a larger performance drop, with Acc-2 decreasing by 0.91% and MAE increasing by 0.017, while removing L_c results in only a 0.51% decrease in Acc-2 and a 0.001 increase in MAE. These findings suggest that L_{uni} facilitates the unimodal subnetworks in

Table 2

Performance comparison on the CMU-MOSI dataset. ALMT* and MSamba* denote the results obtained under the same embedding setting as our model in the same 3 random seeds. MSamba' denotes the results reproduced with the same environment setting reported by the original authors. '+c' denotes the results obtained with context. The superscript 'best' represents the best result under the 3 runs with different random seeds.

Model	CMU-MOSI				
	Acc-2	F1	Acc-7	MAE	Corr
LF-DNN _{ACL18}	77.52/78.63	74.46/78.63	34.52	0.955	0.658
MF _{ICLR19}	77.40/-	77.30/-	34.10	0.965	0.632
LMF _{ACL18}	-/82.50	-/82.40	33.20	0.917	0.695
TFN _{EMNLP17}	-/80.80	-/80.70	34.90	0.901	0.698
MuT _{ACL19}	-/83.00	-/82.80	40.00	0.871	0.698
MISA _{ACM MM20}	81.80/83.40	81.70/83.60	42.30	0.783	0.776
MAG-BERT _{ACL20}	82.13/83.54	81.12/83.58	41.43	0.790	0.766
Self-MM _{AAAI21}	83.44/85.46	83.36/85.43	46.67	0.708	0.796
HyCon _{TAC22}	-/85.20	-/85.10	46.60	0.713	0.790
UniMSE _{EMNLP22}	85.85/86.90	85.83/86.42	48.68	0.691	0.809
ConFEDE _{ACL23}	84.17/85.52	84.13/85.52	42.27	0.742	0.784
DMD _{CVPR23}	-/86.00	-/86.00	45.60	-	-
ALMT _{EMNLP23}	84.55/86.43	84.57/86.47	49.42	0.683	0.805
DBF _{ACL23}	85.10/86.90	85.10/86.90	44.80	0.693	0.801
HyDiscGAN _{IJCAI24}	84.10/86.70	83.70/86.30	43.20	0.749	0.782
MCL-MCF _{TAC24}	84.90/87.30	84.70/87.20	-	0.692	0.799
CLGSI _{NAACL24}	83.97/86.43	83.63/86.25	47.96	0.703	0.790
MSamba _{AAAI25}	85.99/87.43	85.99/87.40	49.67	0.707	0.809
ALMT*	84.65/86.64	84.71/86.64	47.52	0.679	0.806
MSamba*	84.65/86.64	84.71/86.64	47.52	0.679	0.806
MSamba'	82.36/84.60	82.60/84.76	45.19	0.746	0.779
MPFEN_{small}	87.37/89.53	87.28/89.52	50.00	0.582	0.871
MPFEN_{best}_{small}	87.76/89.94	87.68/89.91	49.42	0.600	0.862
MPFEN	87.37/89.48	87.27/89.44	47.67	0.612	0.868
MPFEN_{best}	88.05/90.24	88.00/90.24	48.69	0.589	0.869
MPFEN + c	88.08/90.25	88.00/90.26	50.00	0.586	0.875
MPFEN_{best} + c	88.48/90.70	88.44/90.69	50.44	0.572	0.879

Table 3

Performance comparison on the CMU-MOSEI dataset. ALMT* and MSamba* denote the average results obtained under the same embedding setting as our model in the same 3 random seeds. MSamba' denotes the results reproduced with the same environment setting reported by the original authors. '+c' denotes the results obtained with context. The superscript 'best' represents the best result under the 3 runs with different random seeds.

Model	CMU-MOSEI				
	Acc-2	F1	Acc-7	MAE	Corr
LF-DNN _{ACL18}	80.60/82.74	80.85/82.52	50.83	0.580	0.709
MF _{ICLR19}	78.94/82.86	79.55/82.85	51.34	0.573	0.718
LMF _{ACL18}	80.54/83.48	80.94/83.36	51.59	0.576	0.717
TFN _{EMNLP17}	78.50/81.89	78.96/81.74	51.60	0.573	0.717
MuT _{ACL19}	81.15/84.63	81.56/84.52	52.84	0.559	0.733
MISA _{ACM MM20}	83.60/85.50	83.80/85.30	52.20	0.555	0.756
MAG-BERT _{ACL20}	82.51/84.82	82.77/84.71	50.41	0.583	0.741
Self-MM _{AAAI21}	83.76/85.15	83.82/84.90	53.87	0.531	0.765
HyCon _{TAC22}	-/85.40	-/85.60	52.80	0.601	0.776
UniMSE _{EMNLP22}	85.36/87.50	85.79/87.46	54.39	0.523	0.773
ConFEDE _{ACL23}	81.65/85.82	82.17/85.83	54.86	0.522	0.780
DMD _{CVPR23}	-/86.60	-/86.60	54.60	-	-
ALMT _{EMNLP23}	84.78/86.79	85.19/86.86	54.28	0.526	0.779
DBF _{ACL23}	84.30/86.40	84.80/86.20	54.20	0.523	0.772
HyDiscGAN _{IJCAI24}	81.90/86.30	82.10/86.20	54.40	0.533	0.761
MCL-MCF _{TAC24}	84.20/86.40	84.40/86.30	-	0.536	0.767
CLGSI _{NAACL24}	84.01/86.32	84.21/86.18	54.56	0.532	0.763
MSamba _{AAAI25}	85.78/86.86	85.99/86.93	54.21	0.507	0.796
ALMT*	84.93/87.06	84.88/87.13	56.26	0.501	0.807
MSamba*	85.13/87.53	85.14/87.62	54.66	0.506	0.804
MSamba'	83.32/85.99	83.03/86.10	53.25	0.547	0.766
MPFEN_{small}	85.34/88.09	85.38/87.97	56.65	0.494	0.811
MPFEN_{best}_{small}	85.79/88.06	85.92/87.91	55.87	0.491	0.812
MPFEN	85.23/88.19	85.46/88.11	55.91	0.497	0.809
MPFEN_{best}	85.28/88.44	85.50/88.33	56.47	0.487	0.814
MPFEN + c	85.43/88.75	85.67/88.65	55.50	0.492	0.818
MPFEN_{best} + c	86.16/89.08	86.38/89.00	55.16	0.491	0.815

learning modality-specific emotional cues, which are beneficial to the final cross-modal decision-making. Besides, the incorporation of contextual loss also facilitates the improvement of unimodal subnetworks. This loss encourages each unimodal branch to capture context-aware semantic cues, thereby enhancing its ability to understand modality-specific information in complex scenarios and contributing to a more robust cross-modal fusion.

4.5. Study on the depth of MPFM and MEM

We investigate the impact of different numbers of layers in the Multimodal Progressive Fusion Module (MPFM) and the Multimodal Enhancement Module (MEM) on the model's performance. The depth of both MPFM and MEM is set from 1 to 3.

The experiment results are shown in Table 6. Although different variants with different depth settings achieve competitive performance, we can observe that the (2,2) configuration achieves the best results across most metrics, with Acc-2 and F1 reaching 90.70% and 90.69%, respectively, while MAE decreases to 0.572. This indicates that a moderate number of layers enables the model to more effectively capture cross-modal semantic interactions and enhance information. Therefore, we set the depth of both MPFM and MEM to 2 as default.

4.6. Effect of the unimodal loss coefficient α

To investigate the effect of the unimodal loss coefficient α , we conduct an ablation study by varying the α to study how the model performance change. The experimental results are shown in Fig. 4. We can observe that the Acc-2 metric initially increases with the rise of the unimodal loss coefficient α , reaching its peak at $\alpha = 0.3$. Afterward, Acc-2 exhibits a declining trend. The correlation (Corr) performance remains

Table 4

Performance comparison of different methods in the CH-SIMS dataset. ALMT* and MSamba* denote the results obtained under the same embedding setting as our model in the same 3 random seeds. MSamba* denotes the results reproduced with the same environment setting reported by the original authors. '+c' denotes the results obtained with context. The superscript 'best' represents the best result under the 3 runs with different random seeds.

Model	Acc-5	Acc-3	Acc-2	F1	MAE	Corr
TFN	39.30	65.12	78.38	78.62	0.432	0.591
LMF	40.53	64.68	77.77	77.88	0.441	0.576
MFM	–	–	75.06	75.58	0.477	0.525
MuIT	37.94	64.77	78.56	79.66	0.453	0.564
MISA	–	–	76.54	76.59	0.447	0.563
MAG-BERT	–	–	74.44	71.75	0.492	0.399
Self-MM	41.53	65.47	80.04	80.44	0.425	0.595
ALMT	45.73	68.93	81.19	81.57	0.404	0.619
ConFEDE	46.34	–	81.05	81.13	0.377	0.655
CLGSI	45.95	–	81.18	80.59	0.408	0.634
MSamba	47.17	68.83	82.30	81.75	0.403	0.646
MSamba*	45.54	68.27	80.34	80.49	0.414	0.601
ALMT*	22.10	53.83	70.24	63.64	0.563	0.350
MSamba'	45.51	67.40	80.31	80.31	0.414	0.614
MPFEN _{small}	46.17	71.00	83.26	83.27	0.358	0.707
MPFEN _{small} ^{best}	48.36	71.33	83.59	83.70	0.339	0.734
MPFEN ^{best}	49.67	71.88	84.58	84.77	0.332	0.739
MPFEN ^{best}	48.80	72.43	86.00	86.10	0.339	0.743
MPFEN + c	47.05	70.82	83.15	83.13	0.354	0.713
MPFEN ^{best} + c	46.17	71.77	84.68	84.71	0.345	0.720

Table 5

Ablation study of each component in MPFEN.

Settings	Acc-2	F1	Acc-7	MAE	Corr
w/o MPFM	86.83/88.87	86.76/88.86	47.38	0.644	85.79
w/o MEM	87.61/89.72	87.55/89.73	50.34	0.583	87.34
w/o MDL	88.00/89.89	87.97/89.90	49.03	0.590	87.62
w/o Gating	87.80/89.94	87.74/89.92	50.00	0.581	87.62
w/o L_{uni}	87.51/89.79	87.43/89.76	49.95	0.589	87.10
w/o L_c	88.19/90.19	88.13/90.17	49.18	0.573	87.14
MPFEN	88.48/90.70	88.44/90.69	50.44	0.572	87.90

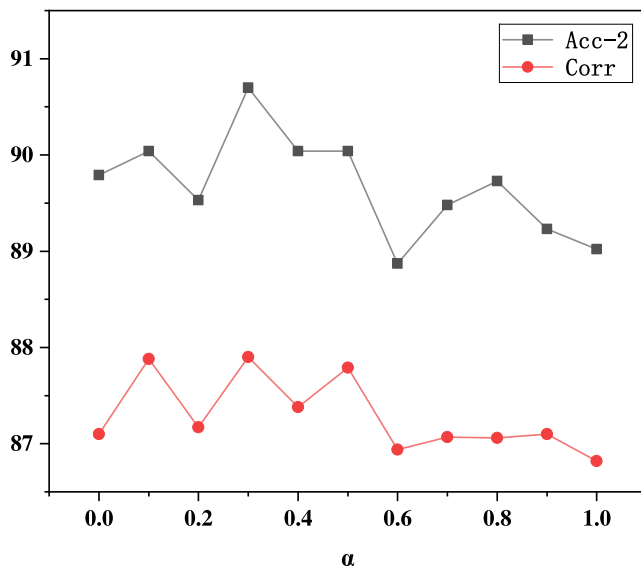


Fig. 4. Performance comparison with different α . The best performance is achieved when the α equal to 0.3.

Table 6

Performance comparison of different layers in CMU-MOSI dataset. The numeral on the left side of the parentheses represents the number of MPFL layers, whereas the one on the right corresponds to the number of MEL layers.

Layer	Acc-2	F1	Acc-7	MAE	Corr
(1,1)	89.56	89.58	49.05	0.595	86.91
(1,2)	89.25	89.23	51.68	0.578	87.09
(2,1)	89.94	89.94	48.18	0.582	87.58
(2,2)	90.70	90.69	50.44	0.572	87.90
(2,3)	90.63	90.62	48.33	0.586	88.19
(3,2)	89.41	89.40	49.20	0.578	87.05
(3,3)	89.71	89.70	49.49	0.584	87.05

Table 7

Ablation of different MPFM architectures on CMU-MOSI. SI-Bi-Mamba denotes the variant where the bidirectional Mamba modules do not share parameters, while Only Cross-Att refers to the setting where Bi-Mamba is removed from the original architecture, and only cross-attention is retained for cross-modal interaction.

Method	Acc-2	F1	Acc-7	MAE
SI-Bi-Mamba	87.75/89.68	87.71/89.67	49.95	0.581
Only Cross-Att	88.00/90.24	87.90/90.20	50.00	0.575
MPFM	88.48/90.70	88.44/90.69	50.44	0.572

relatively high when $\alpha \leq 0.5$. It peaks at $\alpha = 0.3$, but decreases significantly when $\alpha > 0.5$. This suggests that a large unimodal loss coefficient may hinder the effectiveness of multimodal fusion, causing the model to overly focus on optimizing unimodal features, ignoring multimodal interaction and synergy. In summary, appropriately incorporating unimodal auxiliary losses can effectively enhance feature learning quality and improve inter-modal collaboration, whereas excessively strong unimodal supervision may suppress multimodal fusion and degrade overall model performance. Since the model performance achieve peak value when $\alpha = 0.3$, we set $\alpha = 0.3$ as default.

4.7. Study on different MPFM designs

To investigate the effect of different MPFM designs, we study the design of MPFM by exploring two variants: one is with a state-independent Bi-Mamba module (**SI-Bi-Mamba**), where there is no state sharing, and another that relies solely on cross-attention for alignment, referred to as '**Only Cross-Att**'.

The performances of different variants are reported in Table 7. Although SI-Bi-Mamba has more model capacity by removing parameter sharing in Bi-Mamba, and 'Only Cross-Att' employs an effective fusion strategy only with cross-attention, we can observe that both variants, SI-Bi-Mamba and 'Only Cross-Att', show a decline in performance compared to the full MPFM design. This observation suggests that performing local alignment via the shared-parameter Bi-Mamba module prior to global cross-modal attention facilitates the mitigation of fine-grained discrepancies between modalities. Such local alignment reinforces consistency in low-level features across modalities, thereby establishing a stronger foundation for subsequent high-level semantic fusion and ultimately enhancing the overall multimodal integration.

4.8. Investigation on the combinations of different feature encoders

To study the effects of different feature encoders for the MSA task, we conduct an ablation study based on our MPFEN with the different combinations of modal-specific feature encoders on the CMU-MOSI dataset. The combinations of different modal-specific feature encoders includes **BERT-base + Librosa + OpenFace**, **RoBERTa-base + Librosa + OpenFace**, **RoBERTa-large + Librosa + OpenFace**,

Table 8

Comparison of different combinations of unimodal feature encoders in CMU-MOSI. The best is labeled with **bold**.

Combina- tion[Tex- tual + Acous- tic + Visual]	Acc-2	F1	MAE	Acc-7	Corr
BERT-base + Li- broser + Open- Face	84.11/86.74	83.92/86.64	0.714	45.48	0.803
RoBERTa- base + Li- broser + Open- Face	86.73/89.02	86.60/89.01	0.635	47.23	0.845
RoBERTa- large + Li- broser + Open- Face	88.34/90.24	88.27/90.21	0.575	48.98	0.880
RoBERTa- large + Data2vec- base + OpenFace	88.05/89.79	88.02/89.79	0.596	48.10	0.877
RoBERTa- base + Whisper- base + OpenFace	87.03/89.18	86.97/89.17	0.616	48.83	0.859
RoBERTa- large + Whisper- base + TimeS- former	87.90/89.63	87.89/89.66	0.632	47.38	0.870
RoBERTa- large + Whisper- base + OpenFace (Ours)	88.48/90.70	88.44/90.69	0.572	50.44	0.879

RoBERTa-large + Data2vec-base + OpenFace, RoBERTa-base + Whisper-base + OpenFace, RoBERTa-large + Whisper-base + TimeS-former, and RoBERTa-large + Whisper-base + OpenFace (Ours).

The results are shown in Table 8. We can observe that the role of the modal-specific feature extractors is crucial for extracting unimodal sentiment cues for subsequent multimodal learning and fusion. Different encoder combinations have different final performances. The combination **RoBERTa-large + Whisper-base + OpenFace** achieves the best performance compared to other combinations. Thus, we choose the combination **RoBERTa-large + Whisper-base + OpenFace** as the default modal-specific feature encoders in our MPFEN.

In addition, we also transfer the default modal-specific feature encoders in our MPFEN into MSamba* and ALMT* for fair comparison with our MPFEN. As illustrated in Tables 9, experimental results on CMU-MOSI, CMU-MOSEI, and CH-SIMS show that our method consistently outperforms both MSamba* and ALMT* across all 3 datasets. Besides, ALMT* and MSamba* perform worse than their original versions ALMT and MSamba, showing that enhanced feature extractors may not lead to an improvement in performance and a suitable collaborative design of feature extraction and feature fusion is important. It is worth noting that the ALMT* baseline was not optimally tuned for the CH-SIMS Chinese dataset, resulting in suboptimal performance. This observation highlights an important practical consideration: when employing advanced pre-trained feature extractors, directly transferring them to datasets in other languages without careful hyperparameter tuning may lead to performance degradation. Datasets from different linguistic and cultural backgrounds may require distinct optimization strategies. Therefore, researchers should adapt model configurations based on the specific task and dataset characteristics to fully leverage the capacity of powerful feature extractors. These results fully show the effectiveness of the collaborative design of our modal-specific feature encoders, MPFEN, and MEM.

Table 9

The comparison between ALMT, MSamba, and MPFEN under identical embedding settings on the CMU-MOSI, CMU-MOSEI, and CH-SIMS benchmarks.

Model	Dataset	Metrics				
		Acc-2	F1	Acc-7	MAE	Corr
ALMT _{EMNLP23}	MOSI	84.55/86.43	84.57/86.47	49.42	0.683	0.805
MSamba _{AAAI25}		85.99/87.43	85.99/87.40	49.67	0.707	0.809
ALMT*		86.05/88.46	85.88/88.24	43.25	0.721	0.821
MSamba*		84.65/86.64	84.71/86.64	47.52	0.679	0.806
MPFEN _{small}		87.37/89.53	87.28/89.52	50.00	0.582	0.871
MSamba _{AAAI25}	MOSEI	85.78/86.86	85.99/86.93	54.21	0.507	0.796
ALMT _{EMNLP23}		84.78/86.79	85.19/86.86	54.28	0.526	0.779
ALMT*		84.93/87.06	84.88/87.13	56.26	0.501	0.807
MSamba*		85.13/87.53	85.14/87.62	54.66	0.506	0.804
MPFEN _{small}		85.34/88.09	85.38/87.97	56.65	0.494	0.811
ALMT _{EMNLP23}	CH-SIMS	81.19	81.57	–	0.404	0.619
MSamba _{AAAI25}		82.30	81.75	–	0.403	0.646
ALMT*		70.24	63.64	–	0.563	0.350
MSamba*		80.34	80.49	–	0.414	0.601
MPFEN _{small}		83.26	83.27	–	0.358	0.707

Table 10

Further study on MPFEN architecture. In this setting, In **Ours**, MPFEN is configured with one layer. In the att(1–4) variants, MPFEN is set to contain 1 to 4 attention layers, but without the Mamba module.

Model	Acc-2	F1	Acc-7	MAE	Corr	FLOPs
only att(1)	86.88/88.86	86.82/88.91	49.46	0.594	0.869	45.10G
only att(2)	87.27/89.27	87.20/89.19	49.03	0.608	0.869	45.15G
only att(3)	86.68/89.13	86.59/89.09	49.95	0.590	0.866	45.20G
only att(4)	86.96/89.28	86.87/89.26	50.44	0.582	0.873	45.24G
MPFEN _{small}	87.37/89.53	87.28/89.52	50.00	0.582	0.871	45.10G

Table 11

Comparison between attention-based models and MPFL.

Model	Acc-2	F1	Acc-7	MAE	Corr	Parameters	FLOPs
ALMT*	86.05/88.46	85.88/88.24	43.25	0.721	0.821	377.97M	45.66G
MPFEN _{small}	87.37/89.53	87.28/89.52	50.00	0.582	0.871	380.27M	45.10G

4.9. Further study on architectures

To further investigate the differences between our architecture and attention-based structures, we conduct additional experiments by replacing the mamba with attention under a no-context setting on the CMU-MOSI dataset. The results are as shown in the Tables 10 and 11. In Table 10, our model achieves 89.53% in Acc-2 and 0.582 in MAE, surpassing the best pure-attention configuration. It is particularly worth emphasizing that the single-layer hybrid MPFL architecture (Mamba + single-layer attention) achieves better performance than pure-attention variants requiring 3–4 stacked layers, while maintaining lower FLOPs. The efficient implicit alignment provided by Mamba significantly reduces the reliance on multiple attention layers, enabling higher computational efficiency without compromising performance. With relatively low computational cost, our model achieves performance metrics that are on par with or surpass those of multi-layer pure attention mechanisms. This demonstrates that the implicit, sequence-level alignment provided by the Mamba layer substantially reduces the need for repeated attention operations. Simultaneously, we compare our model with the previous attention-based model ALMT, as shown in the Table 11. Although our model has a slightly higher parameter count, it achieves not only lower computational complexity (FLOPs) but also superior performance across all evaluation metrics on the dataset compared to ALMT. These results show the effectiveness of our architecture.

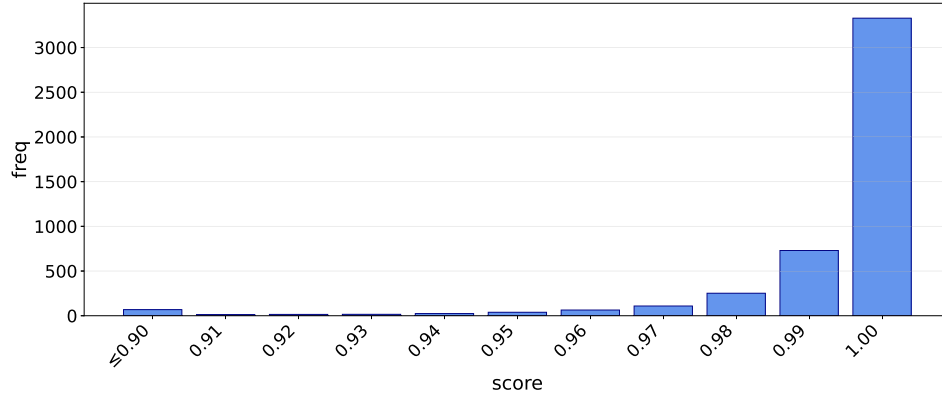


Fig. 5. Visualization of discrepancy scores in VAE on CMU-MOSEI.

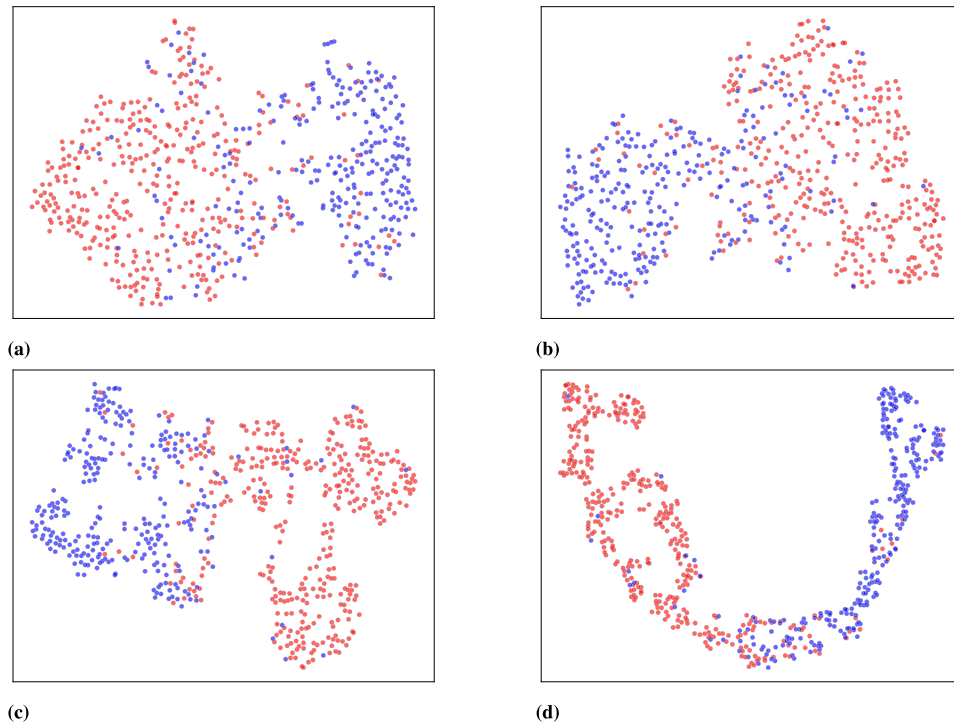


Fig. 6. The t-SNE visualization of the fused features from our MPFEN (c)(d) and the MSamba (a)(b). (a) and (c) show the fused features before the classifier, while (b) and (d) display the feature embeddings from the classifier.

4.10. Visualization of VAE scores

To visualize the discrepancy scores, we conducted experiments on the test set of CMU-MOSEI. The results are shown in Fig. 5, where the sample scores are distributed in the range of 0.57 to 1, where (≤ 0.9) ranges from 0.57 to 0.9. The results show that a considerable portion of the scores are close to 1, indicating that there is a relatively large discrepancy between a substantial amount of multimodal data. In such cases, the model relies on cross-modality-aligned and fused data for sentiment discrimination. Nevertheless, a subset of samples still leverages features from shallowly encoded representations alongside cross-modality-aligned fused features for sentiment identification. This shows the effectiveness of our MDL.

4.11. Feature visualization

To further evaluate the effectiveness of our model in learning discriminative emotional features, we conducted t-SNE visualizations on the test set of the CMU-MOSI dataset in our MPFEN and the MSamba. We visualize both the fused features before the classifier and the feature embeddings from the classifier for comparison.

As shown in Fig. 6, both models are capable of separating positive and negative samples to some extent, demonstrating their ability to capture emotional information. However, compared to our MPFEN, the MSamba model exhibits a noticeably larger overlap between positive and negative samples, indicating more mixed distributions. This suggests that our model has a stronger discriminative power and better feature representation for distinguishing emotional states.

5. Discussion

Recent advances in multimodal intelligence have been largely driven by large language model (LLM)-centric frameworks, such as Flamingo (Alayrac et al., 2022), LLaVA (Liu et al., 2023), and GPT-4V (Yang et al., 2023b). These approaches typically project visual or other modality-specific representations into a shared token space and rely on large-scale Transformer-based language models as the primary engine for cross-modal fusion and reasoning. Owing to their strong sequence modeling and contextual understanding capabilities, such models have demonstrated impressive performance on a wide range of multimodal reasoning tasks. Despite their success, Transformer-centric multimodal reasoning paradigms inherently suffer from quadratic computational complexity. LLM-centric multimodal paradigms, while powerful, are constrained by high computational costs and rigid fusion mechanisms. Meanwhile, state space models like Mamba exhibit notable efficiency in long-sequence modeling. It is therefore valuable to explore a more efficient and effective multimodal fusion framework that combines Mamba's efficient sequence modeling with the strong alignment capability of attention mechanisms.

6. Conclusion

In this work, we propose a novel architecture named Multimodal Progressive Fusion and Enhancement Network (MPFEN) to learn a more discriminative representation for multimodal sentiment analysis. The MPFEN fully explores and models sequential intra-modal information and cross-modal interactions with a hybrid Mamba-Transformer architecture. It consists of three key components, including the Multimodal Progressive Fusion Module (MPFM), the Multimodal Enhancement Module (MEM), and Modality Discrepancy Learning (MDL). The MPFM incorporates state-shared bidirectional Mamba (Bi-Mamba) and cross-modal attention mechanisms to facilitate information fusion across modalities progressively by explicitly modeling centralized cross-modal interaction. The MEM further refines the fused multimodal features with stacked Bi-Mamba-based multimodal enhancement layers. To fully leverage unimodal and multimodal fused features for sentiment prediction, the MDL module is designed to address the heterogeneity among modalities by modeling inter-modal discrepancies and output discrepancy scores that are aggregated with both unimodal and multimodal fused features to construct the final feature representation. Extensive experiments on 3 popular benchmarks, including CMU-MOSI, CMU-MOSEI, and CH-SIMS, demonstrate that our MPFEN achieves superior prediction performance and outperforms state-of-the-art baselines across different benchmarks.

CRedit authorship contribution statement

Jinghui Qin: Conceptualization, Methodology, Software, Validation, Writing – original draft, Writing – review & editing, Funding acquisition; **Zhihao Yang:** Investigation, Data curation, Software, Validation, Visualization, Writing – original draft; **Qite Zhou:** Visualization, Investigation, Data curation; **Lihuang Fang:** Visualization, Investigation, Data curation; **Zhijiang Yang:** Supervision, Writing – review & editing, Funding acquisition.

Data availability

Data will be made available on request.

Declaration of interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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